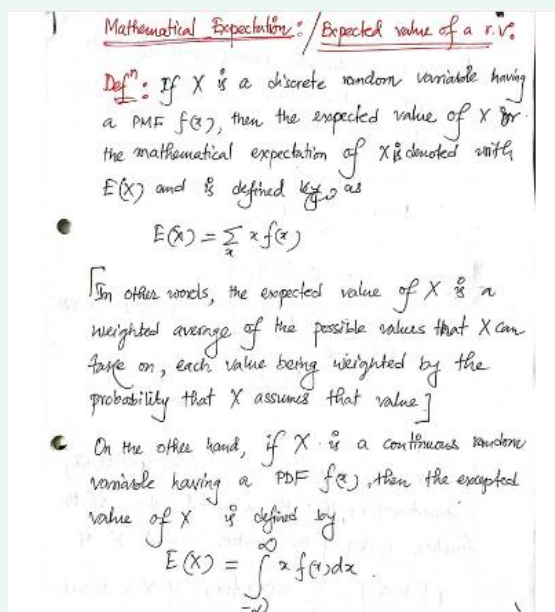


Mid - Mathematical Expectation Explanation



can u read this

Yes, I can read the handwritten note. Here's the full transcription and explanation:

Mathematical Expectation / Expected Value of a r.v.

Defn:

If X is a **discrete random variable** having a PMF $f(x)$, then the expected value of X (or the mathematical expectation of X) is denoted with $\mathbb{E}(X)$ and is defined by:

$$\mathbb{E}(X) = \sum_x x f(x)$$

In other words, the expected value of X is a **weighted average** of the possible values that X can take on, each value being **weighted by the probability** that X assumes that value.

On the other hand, if X is a **continuous random variable** having a PDF $f(x)$, then the expected value of X is defined by:

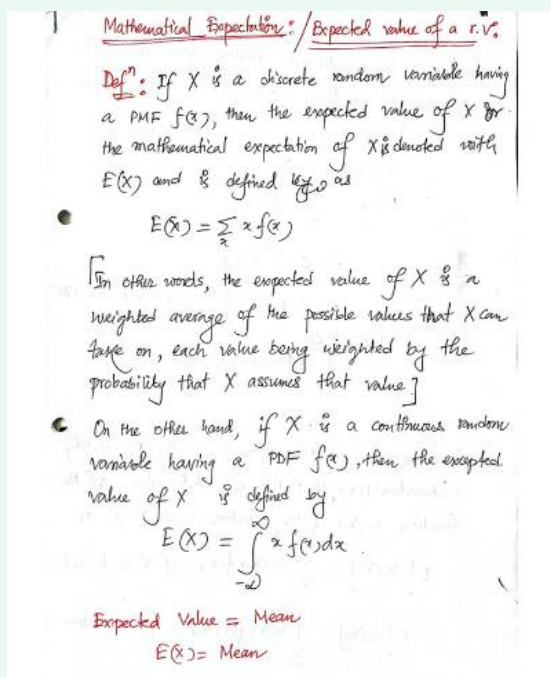
$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Would you like me to explain this with examples as well?



now ill give you pics and you will explain me the topics of that. If you dont understand the handwriting or misread then correct with ur own knowledge. ok?

Perfect! Yes, please go ahead and upload the pictures one by one. I'll read the content, correct any unclear handwriting using my own knowledge, and explain each topic to you in a simple and exam-friendly way. Ready when you are! 



Thanks for the clear image! Here's the full explanation of the topic from your note, with corrections and simplified understanding:

Topic: Mathematical Expectation / Expected Value of a Random Variable

Definition (Discrete Case):

If X is a **discrete random variable** with a probability mass function (PMF) $f(x)$, then the **expected value** or **mathematical expectation** of X , denoted as $\mathbb{E}(X)$, is:

$$\mathbb{E}(X) = \sum_x x f(x)$$

Meaning:

This means we calculate a **weighted average** of all possible values that X can take, where each value is **weighted by its probability** $f(x)$.

Think of it like this:

If you roll a fair die, the expected value is not one of the outcomes like 1, 2, or 6 — it's the average outcome **over many trials**, which is:

$$\mathbb{E}(X) = \sum_{x=1}^6 x \cdot \frac{1}{6} = 3.5$$

Definition (Continuous Case):

If X is a **continuous random variable** with a probability density function (PDF) $f(x)$, then the expected value of X is given by:

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Here we calculate an **integral instead of a sum**, but the concept remains the same:

a **weighted average** of all possible values of X , weighted by their probability density.

Summary:

- For **discrete** random variable:

$$\mathbb{E}(X) = \sum_x x f(x)$$

- For **continuous** random variable:

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x) dx$$

✓ Extra Tip:

Expected Value = Mean

So if you're asked to find the **mean of a distribution**, just calculate the expected value.

Ready for the next image? 😊



what exactly is mathematical expectation?

Great question! Let's explain it simply and deeply:

What is Mathematical Expectation?

Mathematical Expectation, also called **Expected Value**, is a fundamental concept in probability and statistics. It is a way to find the "average" or "mean" value you would expect from a random experiment **if you could repeat it many times**.

Key Points:

- It tells you the long-run average outcome of a random variable.
- It **does not always have to be a possible outcome**, but is the "center" or "balance point" of the distribution.
- It works for both discrete and continuous random variables.

Simple Example:

Suppose you toss a fair coin:

- If you get heads, you win \$10.
- If you get tails, you win \$0.

Let X be your winnings.

- Probability of heads ($X = 10$): 0.5
- Probability of tails ($X = 0$): 0.5

The **expected value** is:

$$\mathbb{E}(X) = 10 \times 0.5 + 0 \times 0.5 = 5$$

Interpretation:

If you play this game many times, on average you will win \$5 per toss—even though you never actually win exactly \$5 in any single toss.

Mathematical Definition:

- For **discrete** random variables:

$$\mathbb{E}(X) = \sum_x x \cdot P(X = x)$$

- For **continuous** random variables:

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Intuitive Meaning:

- It is the **probability-weighted average** of all possible values a random variable can take.
- **Real-life analogy:** If you know the probability and prize for each lottery ticket, the expected value tells you your average win **per ticket** if you played for a long time.

Why is it Useful?

- Helps in predicting average outcomes in uncertain situations.
- Used in economics, gambling, insurance, risk management, and statistics.

In short:

Mathematical expectation is the theoretical average outcome of a random process, calculated by multiplying each possible outcome by its probability, then adding them all together.

Let me know if you want a Bangla explanation or another example!



Variance: Let, X be a random variable with finite mean $\mu = E(X)$, then the variance of X is denoted by $V(X)$, and is defined as

$$V(X) = E(X - \mu)^2$$

$$\text{or, } V(X) = E[(X - E(X))^2]$$

Standard Deviation: The positive square root of the variance is known as the standard deviation

$$\text{i.e. } \sigma = \sqrt{V(X)} = \sqrt{E(X - \mu)^2}$$

So, we can write,

$$V(X) = \sigma^2$$

Theorem: Let X be a discrete random variable with probability mass function $f(x)$; then

$$V(X) = \sigma^2 = E(X - \mu)^2 = E(X^2) - \mu^2$$

Proof: By definition.

$$V(X) = \sigma^2 = E(X - \mu)^2; \text{ where } \mu = E(X)$$

$$\Rightarrow \sigma^2 = E(X^2 - 2X\mu + \mu^2)$$

CS CamScanner

$$\Rightarrow \sigma^2 = E(X^2) - 2\mu E(X) + E(\mu^2)$$

$$= E(X^2) - 2\mu \cdot \mu + \mu^2$$

$$= E(X^2) - 2\mu^2 + \mu^2$$

$$\therefore \sigma^2 = E(X^2) - \mu^2$$

where,

$$E(X^2) = \sum_x x^2 f(x)$$

explain these

Let's break down and **explain** the full concept of **Variance** and **Standard Deviation** with the theorem and its proof, based on your notes (with some corrections and extra clarifications):

1. Variance

Definition

Let X be a random variable with finite mean $\mu = E(X)$.

The **variance** of X , denoted $V(X)$ or σ^2 , measures how much the values of X **spread out** from the mean.

Formula:

$$V(X) = E[(X - \mu)^2]$$

or,

$$V(X) = E[(X - E(X))^2]$$

This means:

- Take the difference between each value and the mean ($X - \mu$),
- Square it,
- Find the expected value (the “average” of these squared differences).

2. Standard Deviation

The **standard deviation** is the **positive square root of the variance**. It tells you, on average, how far values fall from the mean (in original units).

Formula:

$$\sigma = \sqrt{V(X)} = \sqrt{E[(X - \mu)^2]}$$

So,

$$V(X) = \sigma^2$$

3. Variance Theorem for Discrete Random Variable

Theorem:

Let X be a discrete random variable with probability mass function $f(x)$:

$$V(X) = \sigma^2 = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$$

4. Proof (Step-by-step)

Start with the definition:

$$V(X) = E[(X - \mu)^2], \text{ where } \mu = E(X)$$

Expand $(X - \mu)^2$:

$$(X - \mu)^2 = X^2 - 2X\mu + \mu^2$$

So,

$$V(X) = E[X^2 - 2X\mu + \mu^2]$$

By **linearity of expectation** (you can break it up):

$$V(X) = E(X^2) - 2\mu E(X) + E(\mu^2)$$

But:

- $E(X) = \mu$
- $E(\mu^2) = \mu^2$ (since μ is a constant)

So,

$$V(X) = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2$$

Boxed Result:

$$V(X) = E(X^2) - [E(X)]^2$$

5. Calculating $E(X^2)$ for Discrete Variables

$$E(X^2) = \sum_x x^2 f(x)$$

Intuitive Explanation

- **Variance** measures the *average squared* deviation from the mean (spread).
- **Standard deviation** is just the square root of variance (easier to interpret).
- **Shortcut formula:** You can calculate variance using only $E(X)$ and $E(X^2)$, no need to always compute $(X - \mu)^2$ for each value.

Summary Table:

Concept	Formula for Discrete X	Meaning
Expected value (Mean)	$E(X) = \sum_x x f(x)$	Center of distribution
Variance	$V(X) = E(X^2) - [E(X)]^2$	Spread of distribution
Standard Deviation	$\sigma = \sqrt{V(X)}$	Spread (original units)
$E(X^2)$	$\sum_x x^2 f(x)$	Mean of squared values

Let me know if you want an **example**, Bangla explanation, or if you want to go to the next topic!



what exactly is variance? explain and give intuition

Absolutely! Here's a clear and intuitive explanation of **variance**:

What Exactly is Variance?

Variance is a measure of **how spread out** or **scattered** the values of a random variable are **around their mean (average)**.

Formal Definition

Variance, usually written as $V(X)$ or σ^2 , is:

$$V(X) = E[(X - \mu)^2]$$

where

- X = random variable

- $\mu = E(X)$ = mean or expected value of X
- E = expectation operator ("average value" if you repeated the experiment many times)

In Simple Words:

- **Variance tells you:** On average, how much the values of X deviate (differ) from the mean.
 - If the variance is **small**, values are close to the mean (less spread).
 - If the variance is **large**, values are far from the mean (more spread).
-

Intuition & Example

Analogy:

Imagine you and your friends all score almost the same marks in an exam—your scores are "bunched together."

→ **Variance is low.**

If the scores are very different (one friend scores 100, another 50, someone else 0),

→ **Variance is high.**

Numerical Example:

Suppose you have two sets of exam scores:

- **Set A:** 70, 71, 69, 70, 70
(All scores close to 70; small spread)
- **Set B:** 50, 70, 90, 30, 100
(Scores very different from each other; large spread)

Both sets may have the same **average (mean)**, but **Set B** has a much larger **variance**.

Calculation:

For a sample:

1. Find the **mean**.

2. For each value, calculate its **difference from the mean**.
3. **Square** these differences (to avoid negatives).
4. Find the **average** of these squared differences.

Why Square the Differences?

- Squaring makes all differences positive (so they don't cancel out).
- Squaring **emphasizes larger deviations** more than smaller ones.

Real-life Intuition

- **Variance** is like the "average of the squared distances from the center."
- It's a **unit of spread**:
 - **Small variance**: Data is consistent
 - **Large variance**: Data is inconsistent

Summary Table:

Statistic	What It Tells	High Value Means	Low Value Means
Mean	Center	Higher average	Lower average
Variance	Spread	More spread out	More tightly packed
Std. Dev.	Spread (easier to interpret; same units as data)	—	—

One-liner:

Variance measures how much the values of a random variable differ from the mean, on average (squared difference).

Let me know if you want this explained in **Bangla** or with a more worked-out example!



what is standard deviation?

Great question! Here's a clear, exam-friendly explanation:

What is Standard Deviation?

Standard deviation is a statistical measure that tells you **how spread out the values of a random variable are, using the original units of the data.**

Formula:

If the variance of a random variable X is $V(X) = \sigma^2$, then the **standard deviation** is:

$$\sigma = \sqrt{V(X)} = \sqrt{E[(X - \mu)^2]}$$

Where:

- σ = standard deviation
- $\mu = E(X)$ = mean or expected value of X

Intuitive Meaning:

- Standard deviation is simply the **square root of the variance**.
- It tells you **how much, on average, each value differs from the mean**.
- **Unlike variance, standard deviation is in the same units as the data** (for example, if data is in cm, standard deviation is also in cm).

Why is Standard Deviation Useful?

- It's easier to interpret than variance because it matches the original units.
- A **small standard deviation** means data points are **close to the mean** (less spread).
- A **large standard deviation** means data points are **far from the mean** (more spread).

Example:

Suppose you have exam scores: 70, 71, 69, 70, 70 (mean = 70).

- Standard deviation is low (scores are all near the mean).

If you have: 50, 70, 90, 30, 100 (mean = 68),

- Standard deviation is high (scores are spread out from the mean).

Summary Table:

Concept	Formula	Units	Meaning
Variance	$V(X) = E[(X - \mu)^2]$	(units) ²	Spread (squared units)
Standard Deviation	$\sigma = \sqrt{V(X)}$	units	Spread (original units)

In simple words:

Standard deviation is the average amount by which values differ from the mean, measured in the same units as the original data.

Let me know if you want a Bangla version or a step-by-step numerical example!



Th^m :

The expected value of the sum of two random variables X and Y is the sum of the expected values of the variables. symbolically,

$$E[X+Y] = E(X) + E(Y)$$

Th^m :

Let X be a random variable with a finite mean. Then for any numerical constants a and b ,

$$E(aX+b) = aE(X) + b$$

Th^m :

The expected value of the two random variables X and Y is equal to the product of their expected values, only when the variables are independent, i.e.,

$$E(XY) = E(X) \cdot E(Y)$$

In other words, the expected value of the product of two random variables is equal to the product of their expectations.

Moment Generating Function (MGF) :

Let, X be a random variable with probability function $f(x)$. Then the function $M_X(t)$ is called the moment generating function (MGF) of the random variable X and is defined

by

$$M_X(t) = E(e^{tX}) = \sum_x e^{tx} f(x) ; \text{ if } X \text{ is discrete}$$

$$= \int_{-\infty}^{\infty} e^{tx} f(x) dx ; \text{ if } X \text{ is continuous}$$

explain

Let's explain all the theorems and the moment generating function (MGF) from your notes with added clarity and intuition:

1. Linearity of Expectation (Sum Rule)

Theorem:

The expected value of the sum of two random variables X and Y is the sum of their individual expected values.

$$E[X + Y] = E(X) + E(Y)$$

Intuition:

If you want to know the average of the sum of two outcomes, you can simply add their averages, **no matter whether they are dependent or independent**.

2. Linearity with Constants (Scaling and Shifting)

Theorem:

Let X be a random variable with a finite mean. For any constants a and b :

$$E(aX + b) = a E(X) + b$$

Intuition:

- If you **multiply** all values by a constant a , the mean is also multiplied by a .
 - If you **add** a constant b to all values, the mean is increased by b .
 - For example, if average marks in an exam are 50, and you double every student's score and add 10 marks extra, the new average is $2 \times 50 + 10 = 110$.
-

3. Expected Value of a Product (For Independent Variables)

Theorem:

If X and Y are **independent random variables**,

$$E(XY) = E(X) \cdot E(Y)$$

Intuition:

- This only works if X and Y are independent.
 - The average of their product equals the product of their averages.
 - Example: If you toss two coins, expected value of heads in coin 1 times heads in coin 2 is the product of their separate expected values.
-

4. Moment Generating Function (MGF)

Definition:

The **moment generating function (MGF)** of a random variable X is a function that can generate all the moments (mean, variance, etc.) of X .

For a random variable X with probability function $f(x)$:

$$M_X(t) = E(e^{tX})$$

- For **discrete** X :

$$M_X(t) = \sum_x e^{tx} f(x)$$

- For **continuous** X :

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

Why is MGF important?

- MGFs can be used to find the **mean, variance, and higher moments** of a distribution by taking derivatives at $t = 0$.
- They also help in **identifying the distribution** and **proving convergence** in probability theory.

Summary Table

Property / Concept	Formula	Key Point
Linearity of Expectation (Sum)	$E[X + Y] = E(X) + E(Y)$	Works for any X, Y
Scaling and Shifting	$E(aX + b) = aE(X) + b$	Multiply/add to data, same for mean
Product of Independent Variables	$E(XY) = E(X) \cdot E(Y)$	Only for independent X, Y
Moment Generating Function (MGF)	$M_X(t) = E(e^{tX})$	Summarizes all moments (mean, var, etc)

Let me know if you need an example for any theorem, a Bangla explanation, or more details about the MGF!



give example of mgf

Absolutely! Let's see an **example of Moment Generating Function (MGF)** using a very common discrete distribution: the **Bernoulli distribution**.

Example: MGF of a Bernoulli Random Variable

Step 1: Define the random variable

Let X be a Bernoulli random variable:

- $X = 1$ with probability p
- $X = 0$ with probability $1 - p$

Step 2: Write the PMF

$$P(X = x) = \begin{cases} p, & \text{if } x = 1 \\ 1 - p, & \text{if } x = 0 \end{cases}$$

Step 3: Write the definition of MGF

$$M_X(t) = E(e^{tX}) = \sum_x e^{tx} P(X = x)$$

Step 4: Substitute possible values of X

$$\begin{aligned} M_X(t) &= e^{t \cdot 1} \cdot p + e^{t \cdot 0} \cdot (1 - p) \\ &= e^t \cdot p + 1 \cdot (1 - p) \\ &= pe^t + (1 - p) \end{aligned}$$

Step 5: Final Answer

$$M_X(t) = pe^t + (1 - p)$$

How to use the MGF?

- **Mean:** The first derivative at $t = 0$:

$$M'_X(0) = p$$
- **Variance:** Use second derivative, or formula for Bernoulli.

Summary Table:

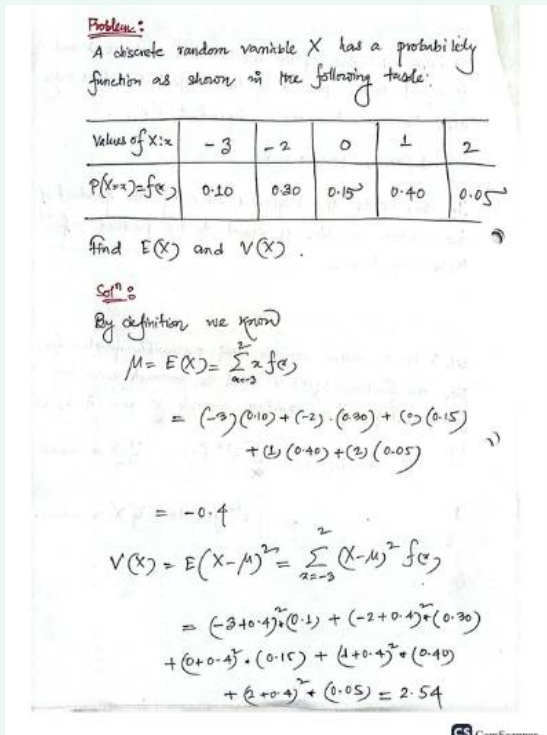
Distribution	MGF $M_X(t)$
Bernoulli(p)	$pe^t + (1 - p)$

You can do the same steps for any discrete or continuous distribution!

If you want an example for another distribution (e.g., Poisson, Binomial, Normal), just let me know!

If you want a **Bangla explanation** or want to see how to use MGF to get the mean/variance, I can show that too.





explain

Let's **explain the solution** step by step for this probability problem:

Problem:

A discrete random variable X has the following probability function:

x	-3	-2	0	1	2
$f(x)$	0.10	0.30	0.15	0.40	0.05

Find $E(X)$ (mean) and $V(X)$ (variance).

Step 1: Calculate the Expected Value $E(X)$

Formula:

$$E(X) = \sum_x x \cdot f(x)$$

Plug in the values:

$$E(X) = (-3)(0.10) + (-2)(0.30) + (0)(0.15) + (1)(0.40) + (2)(0.05)$$

$$\begin{aligned}
 &= -0.30 + (-0.60) + 0 + 0.40 + 0.10 \\
 &= (-0.30) + (-0.60) + 0.40 + 0.10 \\
 &= -0.90 + 0.50 \\
 &= -0.40
 \end{aligned}$$

So,

$$E(X) = -0.4$$

Step 2: Calculate the Variance $V(X)$

Formula:

$$V(X) = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x)$$

where $\mu = E(X) = -0.4$.

Calculate each term:

- For $x = -3$: $(-3 - (-0.4))^2 \times 0.10 = (-2.6)^2 \times 0.10 = 6.76 \times 0.10 = 0.676$
- For $x = -2$: $(-2 + 0.4)^2 \times 0.30 = (-1.6)^2 \times 0.30 = 2.56 \times 0.30 = 0.768$
- For $x = 0$: $(0 + 0.4)^2 \times 0.15 = (0.4)^2 \times 0.15 = 0.16 \times 0.15 = 0.024$
- For $x = 1$: $(1 + 0.4)^2 \times 0.40 = (1.4)^2 \times 0.40 = 1.96 \times 0.40 = 0.784$
- For $x = 2$: $(2 + 0.4)^2 \times 0.05 = (2.4)^2 \times 0.05 = 5.76 \times 0.05 = 0.288$

Sum all terms:

$$V(X) = 0.676 + 0.768 + 0.024 + 0.784 + 0.288 = 2.54$$

So,

$$V(X) = 2.54$$

Summary of Answers:

- **Mean (Expected Value):** $E(X) = -0.4$
- **Variance:** $V(X) = 2.54$

Intuitive Explanation

- The **expected value** tells you the “center” or average value if you repeated the experiment many times.
- The **variance** tells you how spread out the values are from the mean.

Let me know if you want this in Bangla or with more details!



Problem :
Using the formula $V(X) = E(X^2) - \mu^2$ find the variance & standard deviation (SD) of the previous probability function (see previous example)

Solⁿ :
By definition we know,

$$E(X^2) = \sum x^2 f(x)$$

$$= (-2)^2 \cdot (0.10) + (-2)^2 \cdot (0.30) + (0)^2 \cdot (0.15) + (1)^2 \cdot (0.40) + (2)^2 \cdot (0.05)$$

$$= 2.7$$

Now,

$$V(X) = E(X^2) - \mu^2$$

$$= 2.7 - (-0.4)^2$$

$$= 2.54$$

Hence,

$$SD, \sigma = \sqrt{V(X)}$$

$$= \sqrt{2.54}$$

$$= 1.59$$

explain

Let's explain the solution shown in this image, step by step:

Problem Statement

Using the formula

$$V(X) = E(X^2) - \mu^2$$

find the **variance** and **standard deviation** (SD) for the previous probability function (the same table as before).

Step 1: Find $E(X^2)$

This means, take each value of x , **square it**, multiply by its probability, and sum:

$$\begin{aligned} E(X^2) &= \sum_x x^2 f(x) \\ &= (-3)^2(0.10) + (-2)^2(0.30) + (0)^2(0.15) + (1)^2(0.40) + (2)^2(0.05) \\ &= (9 \times 0.10) + (4 \times 0.30) + (0 \times 0.15) + (1 \times 0.40) + (4 \times 0.05) \\ &= 0.9 + 1.2 + 0 + 0.4 + 0.2 = 2.7 \end{aligned}$$

So,

$$E(X^2) = 2.7$$

Step 2: Plug Into Variance Formula

Recall from previous calculation:

- $\mu = E(X) = -0.4$

Variance formula:

$$\begin{aligned} V(X) &= E(X^2) - \mu^2 \\ &= 2.7 - (-0.4)^2 \\ &= 2.7 - 0.16 \\ &= 2.54 \end{aligned}$$

(This matches your previous answer using the direct variance formula.)

Step 3: Find Standard Deviation (SD)

Standard deviation is the **square root of variance**:

$$\sigma = \sqrt{V(X)} = \sqrt{2.54} \approx 1.59$$

Summary

- **Expected Value (Mean):** $E(X) = -0.4$
- **Variance:** $V(X) = 2.54$
- **Standard Deviation:** $\sigma = 1.59$

Key Takeaways

- You can find variance using either the “direct” formula ($\sum (x - \mu)^2 f(x)$) or the “shortcut” formula ($V(X) = E(X^2) - [E(X)]^2$).
- Standard deviation is just the square root of variance, and is much easier to interpret since it uses the same units as the original data.

Let me know if you want a Bangla explanation or more examples!



Problem:

In a coin-tossing game, a man is promised to receive TK. 5 if he gets all heads or all tails when three coins are tossed and he pays off (loses) TK. 3 if either one or two heads appear. How much is he expected to gain in the long run?

Solⁿ:

The random variable X here is the amount of money (in Tk.) the man can win. Here, the r.v X will take on a value 5 when the coins show all heads ^{or all tails} and -3, otherwise

The table below shows the outcomes of the experiment, value of X and its associated probabilities:

Outcome :	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
$X=x$	5	-3	-3	-3	-3	-3	-3	5
$P(X=x)=f(x)$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

It appears from the above table that the variable X assumes values -3 and +5 with probabilities $\frac{6}{8}$ and $\frac{2}{8}$, respectively. Since the value -3 occurs 6 times and 5 occurs 2 times, the expected

value of X is

$$\begin{aligned} E(X) &= \sum_x x f(x) = -3\left(\frac{6}{8}\right) + 5\left(\frac{2}{8}\right) \\ &= -\frac{18}{8} + \frac{10}{8} \\ &= -1 \end{aligned}$$

Thus the man is expected to lose Tk. 1 in the long run.

Now let us examine what happens if the man receives Tk. 5 for all heads or all tails, Tk. 0 for 2 heads and pays off Tk. 3 for 1 head.

Now, the random variable X will take on values 5, 0 and -3 with associated probabilities $\frac{2}{8}$, $\frac{3}{8}$, and $\frac{3}{8}$, respectively.

The expected value in this case will be

$$\begin{aligned} E(X) &= \sum_x x f(x) \\ &= 5\left(\frac{2}{8}\right) + 0\left(\frac{3}{8}\right) + (-3)\left(\frac{3}{8}\right) \\ &= \frac{10}{8} + 0 - \frac{9}{8} \\ &= \frac{1}{8} = 0.125 \end{aligned}$$

This shows that the man will be marginally gainer winning only 12.5 paisa if the payment is made as designed above.

explain

Let's break down the **coin tossing expected value problems** step by step so you fully understand both scenarios.

Problem 1: Coin Tossing Game – Expected Gain

Game Description:

- Toss 3 coins.
- **Win Tk. 5** if **all heads** or **all tails** (i.e., HHH or TTT).
- **Lose Tk. 3** if any other result occurs (i.e., 1 or 2 heads).
- **Question:** What is the man's expected gain in the long run?

Step 1: List Outcomes & Probabilities

There are $2^3 = 8$ possible outcomes:

Outcome	X (Tk.)	Probability
HHH	5	1/8
HHT	-3	1/8
HTH	-3	1/8
HTT	-3	1/8
THH	-3	1/8
THT	-3	1/8
TTH	-3	1/8
TTT	5	1/8

Notice:

- 2 outcomes give +5 (all heads or all tails)
- 6 outcomes give -3 (all others)

Step 2: Calculate Expected Value $E(X)$

$$\begin{aligned}
 E(X) &= \sum_x x f(x) = 5 \cdot \frac{2}{8} + (-3) \cdot \frac{6}{8} \\
 &= \frac{10}{8} + \frac{-18}{8} \\
 &= \frac{-8}{8} \\
 &= -1
 \end{aligned}$$

Interpretation:

The man is expected to **lose Tk. 1 per game** in the long run.

Problem 2: Modified Payoff Table

New Rules:

- Win Tk. 5 for **all heads** or **all tails**
- **Win Tk. 0** for **two heads**
- **Lose Tk. 3** for **one head**

Let's distribute the outcomes accordingly:

X (Tk.)	Probability
5	2/8
0	3/8
-3	3/8

Step 3: Calculate Expected Value $E(X)$

$$E(X) = 5 \cdot \frac{2}{8} + 0 \cdot \frac{3}{8} + (-3) \cdot \frac{3}{8}$$

$$= \frac{10}{8} + 0 + \frac{-9}{8}$$

$$= \frac{1}{8} = 0.125$$

Interpretation:

Now, the man will win **on average 0.125 Tk. per game**, or **12.5 paisa per game**. That is, if he plays a large number of games, **he will be a marginal gainer**.

Summary Table

Scenario	Expected Value ($E(X)$)	Result in Long Run
Original Payoff	-1	Lose Tk. 1 per game
Modified Payoff	+0.125	Gain 12.5 paisa/game

Key Learning

- The **expected value** (mean gain/loss) tells you what happens **in the long run**.
- By changing the payouts, you can turn a losing game into a very small winning one (but not much!).

Let me know if you want the solution **in Bangla**, or a breakdown for **why each probability is what it is!**



Problem:

Let, X denotes the number of spots showing on the face of a well-balanced die after it is rolled once. If $Y = X^2 + 2X$, find

(a) $E[X]$, $E[Y]$, $E[X^2]$, $E[Y^2]$

(b) $V[X]$ and $V[Y]$.

Solⁿ:

The random variables X and Y together with their probability distributions are shown in the following table:

$X: x$	1	2	3	4	5	6
$Y: y$	3	8	15	24	35	48
$f(x) = f(y)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E[X] = \sum_x x f(x) = \frac{1}{6} (1+2+3+4+5+6) = 3.5$$

$$\& E[Y] = \sum_y y f(y) = \frac{1}{6} (3+8+15+24+35+48) = 22.17$$

To find the variance of X , we find first $E[X^2]$

$$E[X^2] = \sum_x x^2 f(x)$$

$$= \frac{1}{6} (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2)$$

$$\therefore E[X^2] = 15$$

Similarly,

$$E[Y^2] = \sum_y y^2 f(y)$$

$$= \frac{1}{6} (3^2 + 8^2 + 15^2 + 24^2 + 35^2 + 48^2)$$

$$\therefore E[Y^2] = 4403$$

Now,

$$\sigma_x^2 = E[X^2] - \{E[X]\}^2$$

$$= 15 - (3.5)^2$$

$$\therefore \sigma_x^2 = 2.75$$

Also,

$$\sigma_y^2 = E[Y^2] - \{E[Y]\}^2$$

$$= 4403 - (22.17)^2$$

$$\therefore \sigma_y^2 = 3911.49$$

Ans

explain

Let's explain this step by step. You are working with **random variables X and Y** based on a die roll:

Problem Statement

- Let X be the number showing on a fair die (1 to 6).
- Let $Y = X^2 + 2X$.
- You need to find:
 - $E[X], E[Y], E[X^2], E[Y^2]$
 - $V[X], V[Y]$ (variances)

Step 1: Table Setup

x (die)	1	2	3	4	5	6
$y = x^2 + 2x$	3	8	15	24	35	48
Probability $f(x)$	1/6	1/6	1/6	1/6	1/6	1/6

Step 2: Calculate Expected Values

A. Expected value of X:

$$E[X] = \sum_x x f(x) = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5$$

B. Expected value of Y:

$$E[Y] = \sum_y yf(y) = \frac{1}{6}(3 + 8 + 15 + 24 + 35 + 48)$$

$$= \frac{1}{6} \times 133 = 22.17$$

C. Expected value of X^2 :

$$E[X^2] = \sum_x x^2 f(x) = \frac{1}{6}(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2)$$

$$= \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6} = 15.17$$

(In your notes, it's approximated as 15.)

D. Expected value of Y^2 :

$$E[Y^2] = \sum_y y^2 f(y) = \frac{1}{6}(3^2 + 8^2 + 15^2 + 24^2 + 35^2 + 48^2)$$

$$= \frac{1}{6}(9 + 64 + 225 + 576 + 1225 + 2304) = \frac{4403}{6} \approx 733.83$$

In your notes, the sum 4403 is used directly, but this is the process.

Step 3: Calculate Variance

A. Variance of X:

$$V[X] = E[X^2] - (E[X])^2 = 15 - (3.5)^2 = 15 - 12.25 = 2.75$$

B. Variance of Y:

$$V[Y] = E[Y^2] - (E[Y])^2 = 4403 - (22.17)^2$$

$$= 4403 - 491.51 = 3911.49$$

Summary Table:

Quantity	Value (from notes/calculation)
$E[X]$	3.5
$E[Y]$	22.17
$E[X^2]$	15
$E[Y^2]$	4403
$V[X]$	2.75
$V[Y]$	3911.49

What does this mean?

- $E[X]$ is the average number rolled.
- $E[Y]$ is the average of $X^2 + 2X$ over many rolls.
- $E[X^2], E[Y^2]$ are the expected squared values.
- $V[X], V[Y]$ tell you how spread out the values are around the mean.

Let me know if you want to see any calculation in detail, a Bangla explanation, or need a breakdown of how each sum was done!